

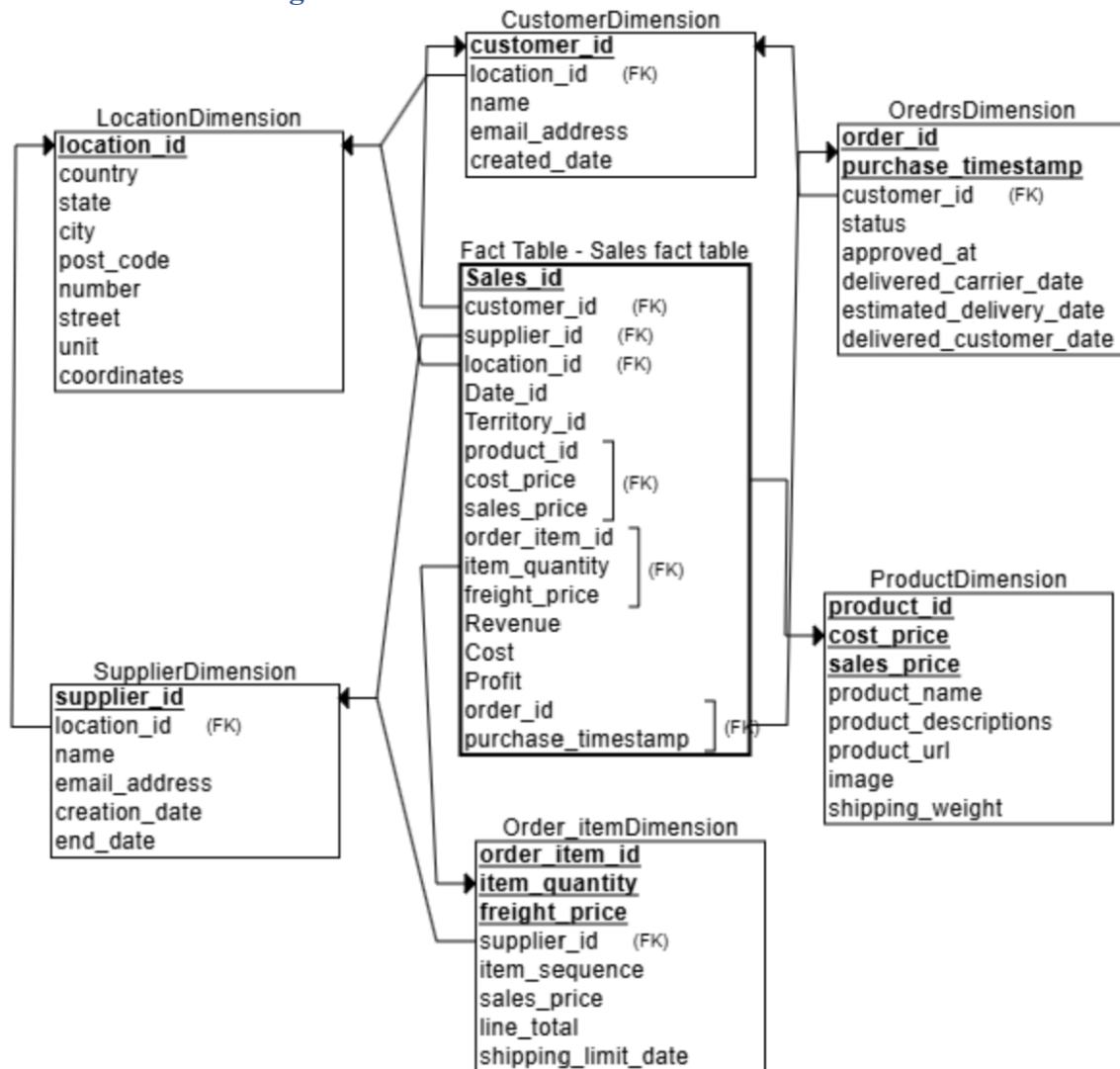
Assessment 1

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1. Dimensional model (6 Marks):

You are required to design a star schema and describe and justify it. In the report, include a diagram of your star schema^[1], justifications for the design choices that you make, and a short description of data grains, dimensions, and fact measures.

1.1 Dimension Model Image



[1] You can use ERDPlus (<https://erdplus.com/>) online or draw your design using MS Word or other tools.

1.2 Description of Dimensional Model (No more than 500 Words):

Kimball's four-step process

1. Choose the Business Process: The core business process selected is sales transactions through Amazon's portal. Each sale of a product to a customer forms the foundation for analysis, matching Crocoroos's goal to track revenues, customer behavior, supplier performance, and product profitability.

2. Declare the Grain: The grain of the fact table is set at the individual order item level. Each record in the fact table represents one line item within a customer's order, capturing sales revenue, cost, freight charges, and profit for that specific product sale.

3. Identify the Dimensions: The dimensions identified are:

CustomerDimension: Customer attributes (location, name, email)

ProductDimension: Product details (cost price, sales price, product description)

SupplierDimension: Supplier information (name, email, linked location)

Order_itemDimension: Order item-specific data (item quantity, freight price)

LocationDimension: Geographic attributes (country, state, city)

OrdersDimension: Order-level information (purchase timestamps, shipping status)

4. Identify the Facts

The primary facts recorded are:

item_quantity: Number of units sold

sales_price: Selling price per unit

cost_price: Cost per unit from the supplier

freight_price: Shipping cost of the item

Revenue: Calculated as $\text{item_quantity} \times \text{sales_price}$

Cost: Calculated as $\text{item_quantity} \times \text{cost_price}$

Profit: Derived as Revenue - Cost

Design Justification:

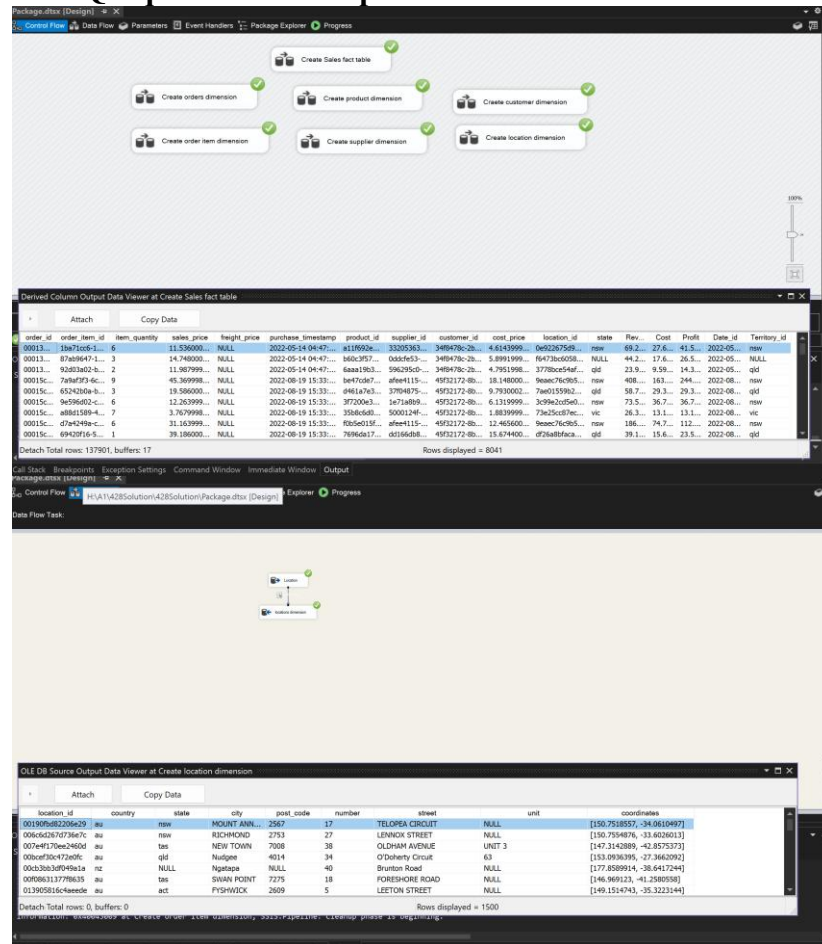
The dimensional model using a star schema was specifically designed to support Crocoroos's operational challenges and strategic goals as an online retailer. Given that Crocoroos (CORS) relies heavily on historical sales to predict demand, a fast and flexible analytics framework is critical for success. It allows the company to track revenue, cost, freight, and profitability at the most granular transaction level. Data warehouse is essential for the company to accurately identify high-demand and high-margin products and can also forecast future product trends more reliably based on actual customer behavior rather than broad estimates.

Linking to a CustomerDimension and LocationDimension enables insights into key markets, top customers (such as identifying the five biggest Australian customers), and regional sales performance (such as revenue from New Zealand). By tracking SupplierDimension, Crocoroos can evaluate supplier efficiency and cost structures, which is vital for a business that relies on bulk purchases to minimize costs. In short, this star schema supports their immediate need for understanding revenue sources, key customers, and geographical sales patterns, while also providing a foundation for longer-term strategic planning such as supply chain optimization and margin improvement initiatives. The dimensional model offers simplicity for end-users, powerful performance for business analytics, and enough flexibility to adapt as the company's needs evolve.

2. Implementation of the Dimensional Model:

You are required to implement the dimensional model using the SQL Server Integration Services (SSIS) to integrate and transform data. In the report, add a screenshot of your SSIS transformations and briefly explain your transformations and steps.

2.1 Dimensional Tables implementation (Images from ETL data flow design: One image of data flow process for dimension table. Please use the tasks from SSSIS toolbox and don't use SQL queries. An exception can be made for date dimension).



Package.dtsx [Design] | Control Flow | Data Flow | Parameters | Event Handlers | Package Explorer | Progress

Data Flow Task:

OLE DB Source Output Data Viewer at Create customer dimension

customer_id	location_id	name	email_address	created_date
4f6b2ba-36ce-b0f6-8c59-a...	51012a37d6d51a53	Derrick Peters	petersen@brian@example.com	2021-10-28 09:20:45...
28853ba3-32a7-4f6c-9126-0...	dec55c708b8c4a98	Jennifer Garrett	hodge@sonya@example.com	2021-10-28 09:22:02...
68a8e93-a7d9-4ecc-99d4-6...	2a6e571f52477ce4	Gabriel Moore	alexanderpau@example.net	2021-10-28 09:47:28...
080701b-081-4d4b-95f4-df...	8a72951a603143	Lauren Whitehead	dewar5@example.com	2021-10-28 10:06:15...
3d7fa13a-7f66-46de-ac29-e...	2a6c4537494000ce	Anthony Hernandez	padlakrist@example.org	2021-10-28 10:19:39...
3e8f6a8d-1a81-4911-99d8-...	8253c19a6d7f5c71	John Beard	sanaperez@example.net	2021-10-28 10:26:44...
3502903a-99f4-474b-8a7a-...	73ba3d51b3333d	Sarah Foster	green@rudy@example.net	2021-10-28 11:10:49...

Attach Total rows: 0 buffers: 0 Rows displayed: 9916

Package.dtsx [Design] | Control Flow | Data Flow | Parameters | Event Handlers | Package Explorer | Progress

Data Flow Task:

OLE DB Source Output Data Viewer at Create supplier dimension

supplier_id	location_id	name	email_address	creation...	end_date
0156d809-052a-43a5-8b09-...	a4d189f29f70361	Robert Sims	rosen@2@example.net	2022-09...	None
029e1396-7878-4fbc-9f7c-c5...	206321a94528fa4a	Alex Franklin	yam@strong@example.com	2023-02...	None
038ae07-f6c4-4a6f-887a-8...	d7e339a433dbdab	Stacey Phillips	kirk76@example.com	2021-12...	8/8/2022 18:50
0a6d1493-f5c4-4a23-69f6-8...	62a4933ba657f8f	Adam Underwood	prince@j@example.org	2021-12...	None
0ddcd53-62ba-4767-aad7-f...	86473b00580202	Alexander Wagner	jasonwoods@example.com	2021-12...	26/1/2022 1:46
123a973-a100-433c-aad9-...	09a80f46b8b187f	Jessica Spears	neal@uth@example.org	2021-11...	None
1284a09-7d07-4927-aad4-...	7773ba55a4209a	Albert Rhodes	h@mail@example.org	2023-01...	None

Attach Total rows: 0 buffers: 0 Rows displayed: 100

Call Stack | Breakpoints | Exception Settings | Command Window | Immediate Window | Output

Package.dttax [Design] | Control Flow | Data Flow | Parameters | Event Handlers | Package Explorer | Progress

Data Flow Task:

OLE DB Source Output Data Viewer at Create orders dimension

order_id	customer_id	delivered_order_date	estimate	delivered_customer_date	purchase	approved	status
00013423-464e-4869-4869-...	348476c-2b2e-4b3e-92be-5...	2022-05...	NULL	2022-05...	2022-05...	shipped	
00015c0b-d624-48c3-9e41-5...	45f32172-8b0d-4399-9660-...	2022-08...	NULL	2022-08...	2022-08...	invoiced	
00021212-4e1a-41c8-5176f-5...	ed0d8f9f-5c0e-4f2a-bc17-3...	2023-03...	NULL	2023-03...	NULL	created	
00041333-4990-4403-9fec-f...	d7634d86-df93-4ed6-5167-f...	2023-03...	NULL	2023-03...	2023-03...	delivered	
00048f9b-bc25-454d-4723-9...	4d8e1f8b-494e-4462-4872-0...	2023-03...	NULL	2023-03...	NULL	processing	
00071a4f-9995-460d-9023-...	765d23e-807f-4223-8223-...	2023-03...	NULL	2023-03...	NULL	created	
00096b32-4661-4ee3-e720-1...	ca0c476-c267-4204-afce-1...	2023-01...	NULL	2022-12...	2022-12...	created	

Detach Total rows: 25000, buffers: 2 Rows displayed = 9994

Package.dttax [Design] | Control Flow | Data Flow | Parameters | Event Handlers | Package Explorer | Progress

Data Flow Task:

OLE DB Source Output Data Viewer at Create product dimension

product_id	product_name	product_descriptions	shipping_weight	image	product_url	product_subcategory_id	cost_price	sales_price
000590576...	Dance 1177...	<Long Text>	1.6 ounces	https://images-na.ssl-image...	https://www.amazon.com/D...	PS_5	4.4850001...	8.9700002...
001a6d12e...	AMERICAN G...	<Long Text>	1.4 ounces	https://images-na.ssl-image...	https://www.amazon.com/A...	PS_36	5.6951999...	14.237999...
002a020fb...	Forum Novel...	<Long Text>	6.4 ounces	https://images-na.ssl-image...	https://www.amazon.com/F...	PS_7	4.3680000...	8.7360000...
005410480...	Aurora World...	<Long Text>	7.2 ounces	https://images-na.ssl-image...	https://www.amazon.com/A...	PS_2	9.6000003...	19.200000...
006023035...	Rule's Man...	<Long Text>	11.2 ounces	https://images-na.ssl-image...	https://www.amazon.com/R...	PS_19	5.2000001...	13.020000...
006646439...	Dimeco's Pita...	<Long Text>	4.8 ounces	https://images-na.ssl-image...	https://www.amazon.com/D...	PS_13	2.4594000...	6.1459999...
009b1672fe...	DC Collectib...	<Long Text>	8 ounces	https://images-na.ssl-image...	https://www.amazon.com/D...	PS_26	17.639999...	35.279998...

Detach Total rows: 2199, buffers: 2 Rows displayed = 956

Package.dttax [Design] | Control Flow | Data Flow | Parameters | Event Handlers | Package Explorer | Progress

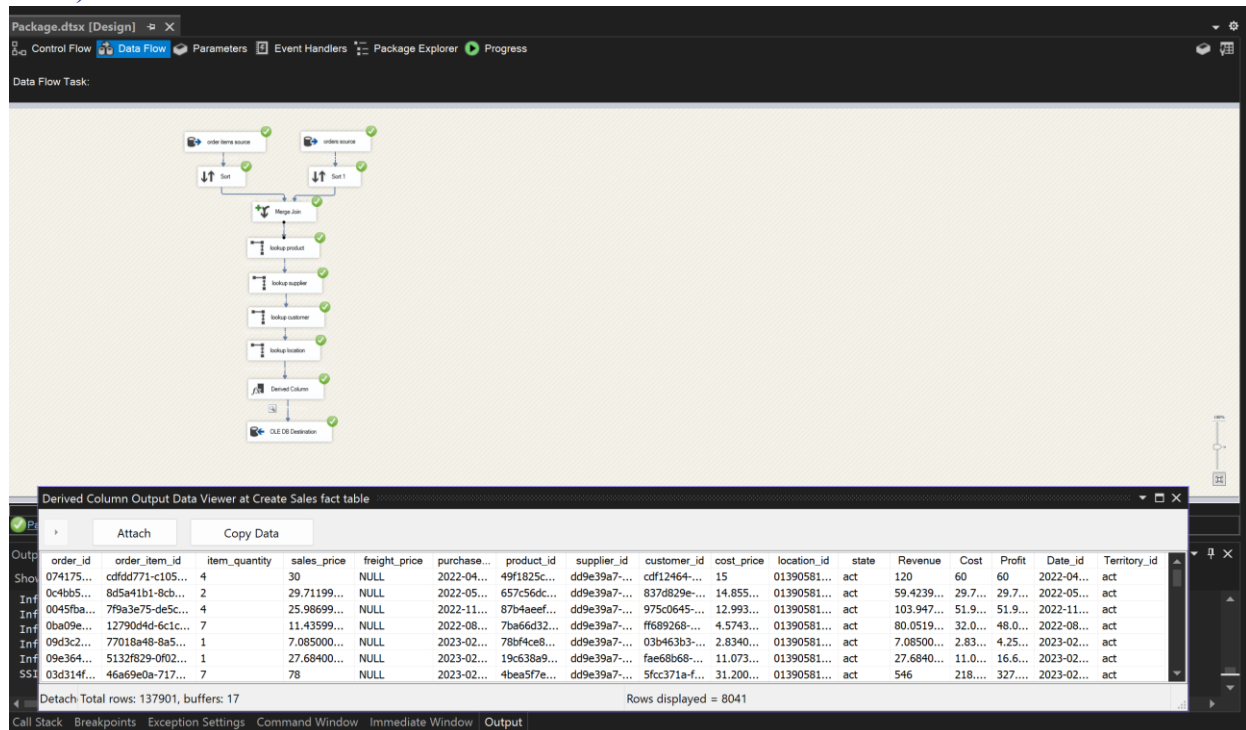
Data Flow Task:

OLE DB Source Output Data Viewer at Create order item dimension

order_item_id	order_id	product_id	supplier_id	item_sequence	item_quantity	sales_price	line_total	weight_price	shipping_limit_date
0a398091-4477-4...	20062c0d...	0a724651de430032f86892...	27070baa-4702-4471-...	0	13.20599...	79.7700000785625	NULL	2023-03-28 04:24:07...	
11126240-96a9-4...	3113366...	869779510e05654e54a59b...	b7a5572b-6fcb-460d-...	0	2	103.5449...	207.089996317891	NULL	2023-03-18 01:40:09...
000d0c0f-f0c0-3...	6895a09...	409774e20c9733809318679e...	e71a8b9-75af-42ab-...	0	1	24.61199...	24.6119995117188	NULL	2022-11-18 15:39:37...
0811140a-950a-4...	2728791...	51a313a0d0404e510791431...	d5f42a2a-af0e-441b-...	0	4	9.631099...	38.54399871832617	NULL	2023-01-02 16:58:41...
030a6865-a2ee-4f...	b67c0db...	3481595e40b046c55ef4c1...	afee1115-dfaa-45a3-...	0	6	5.940000...	35.6399993896404	NULL	2023-03-12 03:37:04...
08117935-831e-4...	4a84e23...	869a8e799219765a0a80dc...	0a0d1493-f5cb-4e29-...	0	8	8.949999...	46.799992370605	NULL	2022-10-27 18:26:38...
08172735-c965-4...	cbaf0f9...	088f7c46913a686c39512...	3f6a23ca-941c-404e-8...	0	9	472.1470...	4249.3212421675	NULL	2022-08-18 14:37:20...

Detach Total rows: 137901, buffers: 13 Rows displayed = 9909

2.2 Fact Table implementation (Image from ETL data flow design: one image from the fact table)



2.3 Description of the Implementation process (No more than 500 Words)

In this project, the dimensional model was implemented using SQL Server Integration Services (SSIS).

First, we need to first restore the data of Sales and then create the dimension tables. The six dimension tables were created individually. For each dimension (Customer, Supplier, Product, Location, Order, Order Item), a OLE DB Source was used to extract the raw data and then create the dimension table with the surrogate keys. The cleaned data was loaded into each dimension table through an OLE DB Destination.

To build the Sales Fact Table, two data sources (Order_Items and Orders) were extracted and Sorted by order_id, then combined using a Merge Join component. After merging, multiple Lookup transformations were used to retrieve the surrogate keys from each dimension table, based on business keys like product_id, supplier_id, and customer_id.

A Derived Column was used to calculate key business metrics:

Revenue = item_quantity × sales_price

Cost = item_quantity × cost_price

Profit = Revenue - Cost

Date_id = time_stamp

Territory_id = state

Finally, all transformed data was mapped properly into the SalesFact table using OLE DB Destination with fast load mode. Throughout the process, Data Viewers were used at key connection points to verify data integrity and row counts before loading into the database.

This ETL process successfully created a clean star schema, connected fact and dimension tables properly, handled missing data gracefully, and ensured efficient loading for future reporting and analysis.

3. Analytical Queries (6 Marks)

Once the dimension model is implemented, you can run the SQL queries to generate business insights for Crocoroos. The insights should answer questions with respect to key customers and most profitable products and sales territories as follows: You are required to query the dimensional model to answer the analytical queries. Please run SQL queries to answer the analytical questions and report the screenshot of the outputs of your query and the SQL statement.

3.1 Who are the five biggest customers in Australia:

SQL query:

```
SELECT TOP 5
    sf.customer_id,
    SUM(sf.revenue) AS total_revenue
FROM
    dbo.SalesFactTable sf
JOIN
    [dbo].[locations dimension] ld
ON
    sf.locationskey = ld.locationskey
WHERE
    ld.country = 'au'
GROUP BY
    sf.customer_id
ORDER BY
    total_revenue DESC;
```

Result from the above query:

	customer_id	total_revenue
1	5644e614-2512-41a0-810b-652e1a9d26dc	107218.898866177
2	79b2122b-4799-4715-9252-d94c24f1cce9	98264.8954153061
3	279e595c-f8e9-4918-8b75-439b198e382b	88139.9868273735
4	406a8bb9-02c1-426f-bfe1-c5c269d7f731	82219.2546653748
5	c772f14b-fdfe-49ec-b390-12205887c2d6	79277.1757054329

3.2 How much is the total monthly revenue from New Zealand?

SQL query:

```
SELECT
    FORMAT(purchase_timestamp, 'yyyy-MM') AS month_year,
    SUM(revenue) AS total_monthly_revenue
```

```

FROM
    dbo.SalesFactTable sf
JOIN
    [dbo].[locations dimension] ld
ON
    sf.locationskey = ld.locationskey
WHERE
    ld.country = 'nz'
GROUP BY
    FORMAT(purchase_timestamp, 'yyyy-MM')
ORDER BY
    month_year;

```

Result from the above query:

	month_year	total_monthly_revenue
1	2021-11	46450.4497981071
2	2021-12	143191.454553604
3	2022-01	266729.714894295
4	2022-02	429892.828960493
5	2022-03	484337.580932993
6	2022-04	529505.666224957
7	2022-05	723027.600790262
8	2022-06	996934.312856197
9	2022-07	1249118.27480674
10	2022-08	1267029.66990352
11	2022-09	1502531.93219185
12	2022-10	1919351.81526818
13	2022-11	2300667.80806661
14	2022-12	2792553.15167069
15	2023-01	3867277.68494002
16	2023-02	4677196.5883703
17	2023-03	2829505.51032782

3.3 What are the most profitable territories over the year?

SQL query:

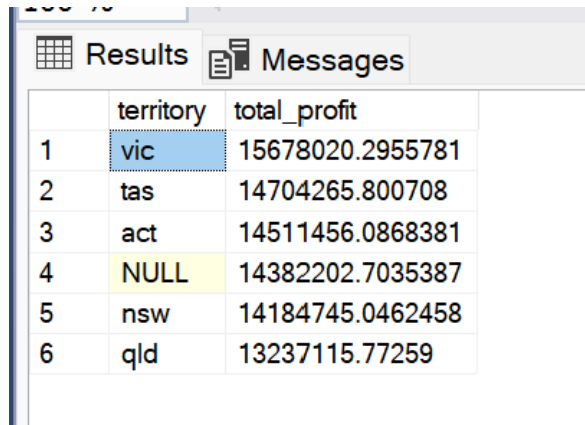
```

SELECT
    state AS territory,
    SUM(profit) AS total_profit
FROM
    dbo.SalesFactTable sf
JOIN
    [dbo].[locations dimension] ld
ON
    sf.locationskey = ld.locationskey
GROUP BY
    state

```

ORDER BY
total_profit DESC;

Result from the above query:



The screenshot shows a database interface with two tabs: 'Results' and 'Messages'. The 'Results' tab is active, displaying a table with two columns: 'territory' and 'total_profit'. The table contains six rows of data, ordered by 'total_profit' in descending order. The first row is highlighted in blue, and the fourth row is highlighted in yellow.

	territory	total_profit
1	vic	15678020.2955781
2	tas	14704265.800708
3	act	14511456.0868381
4	NULL	14382202.7035387
5	nsw	14184745.0462458
6	qld	13237115.77259